Sketching Rational graphs

Key Points

To sketch $f(x) = \frac{N(x)}{D(x)}$

- We find the y-intercept from $f(0)$ and the x-intercepts by solving $N(x) = 0$
- We calculate the undefined asymptotes by solving $D(x) = 0$. For which we look at the value either side of the asymptote
- We find limits as $x \to \pm\infty$ (Note graphs may cross the limits)
- We find Min/Max by calculus (if required)

Basic Skills

Q1. Sketch the graph of $y = \frac{1}{5x+2}$

Q2. Find the following for $y = \frac{2x-5}{5x+3}$

a) When does $y = 0$

b) What is y when $x = 0$

c) At what value of x is y undefinded

d) At either side of the undefined value what happens to y

e) Limit of y as $x \to \infty$ and $x \to -\infty$

f) Hence sketch the graph of y

Q3. Sketch the graph of $y = \frac{6x}{8-3x}$

Q4. Sketch the graph of $y = \frac{1}{(x-1)(x+2)}$

Competency Skills

Q5. Find the following for graph of $y = \frac{4x+3}{x^2+5x-24}$

a) The intercepts

b) The vertical (undefined) asymptotes and the behaviour at either side

c) The limits as $x \to \pm\infty$

d) Hence plot the graph

Q6. Sketch the graph of $y = \frac{x(x-1)}{(x-2)(x+3)}$ showing any minimums and maximums

Q7. By sketching two appropriate graphs solve the inequality $\frac{x+5}{4x-2} \geq 2$

Q8. **a)** Sketch the graph of $y = \frac{x^2+x-6}{x^2-x-6}$

b) Hence solve the inequality $\frac{x^2+x-6}{x^2-x-6} < \frac{1}{2}$

Advanced Skills

Q9. For the graph of $y = \frac{x^2-2x-5}{x^3+x^2-17x+15}$

a) Given that 1 is a root of $x^3 + x^2 - 17x + 15$ find the vertical asymptotes

b) Find the intercepts

c) Find the limits of the graph as $x \to \pm\infty$

d) Hence plot the graph

Q10. By sketching the graph of $y = \frac{(x-5)(2x+3)}{(x-1)^2}$ state what behaviour happens when the denominator has repeated roots

Q11. What are the equations of the vertical asmpytotes of $y = \frac{x^n-a^n}{x-a}$ for any $a \in \mathbb{R}$ and any $n \in \mathbb{N}$

Extension

Q12. The graph $y = \frac{x^2+2}{2x+5}$ goes to infinity as $x \to \infty$ but under what line does it tend to infinity

Hint: Polynomial division to find the leading term